

LOCAL LLMs

My 8GB VRAM vs the World

From centralized AI to models you can own

Presenter · Organization · Date



A PRACTICAL STORY OF
OWNERSHIP, CONSTRAINTS,
AND CONTROL

Your machine. Your model. Your move.

Stockfish proposes credible moves.
Qwen3-4B chooses one through
llama.cpp—and explains it without
sending the game away.

Qwen3-4B

Q4_K_M

RTX 4060

Local only

LOCAL DECISION

Candidate moves

- 1 e4
- 2 Nf3
- 3 d4

"I'll take space in the center while keeping options open."

LOCALHOST

Two useful tiers—not one winner

CLOUD

Hosted frontier

- Maximum general capability
- Large-scale tools and context
- Someone else runs the infrastructure

LOCAL

Local / open-weight

- Private and offline workflows
- Control over model and runtime
- Fast, bounded, repeatable tasks

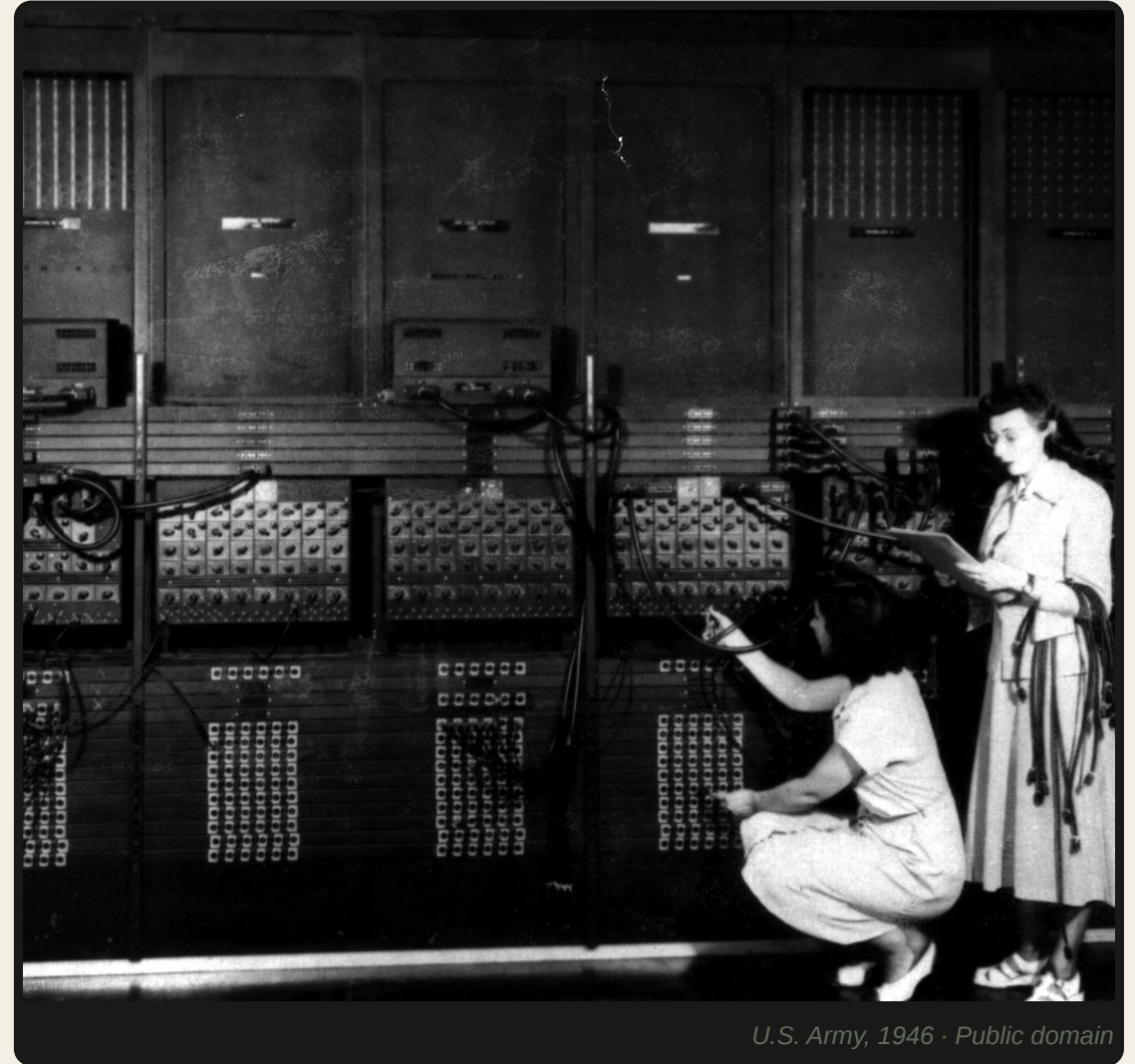
Local complements cloud. The job decides the tool.

Capability arrived before accessibility

ENIAC filled a room. Programming it meant physically reconfiguring the machine.

ROOM-SCALE

Centralized capability came first.



U.S. Army, 1946 · Public domain



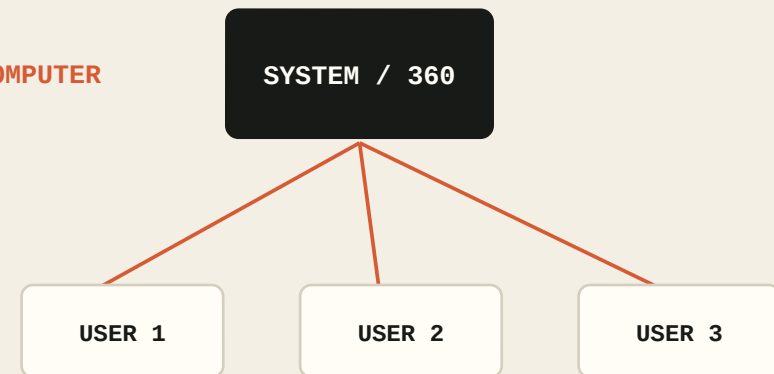
Warren K. Leffler / Library of Congress, 1967 · Public domain

THE MAINFRAME ERA

One powerful machine. Many users.

Access was shared, scheduled, and mediated by an institution.

CENTRAL COMPUTER



Many access points · one shared center

THE PERSONAL-COMPUTER SHIFT

Ownership changed who could experiment

The Kenbak-1 was one of the earliest commercial personal computers. The revolution was not only smaller hardware—it was direct access.

DIRECT ACCESS

Personal access changed the pace of learning.



Kathryn Greenhill · CC BY-SA 2.0

Decades in computing. Years in language models.

COMPUTING



LANGUAGE MODEL



A lens, not a prediction: personal and centralized computing coexist.

A new layer of personal computing

01

Privacy

Inputs can stay on-device

02

Control

Choose the model, version, and rules

03

Offline

Work after the model is downloaded

04

Latency

No network round trip

05

Cost

No per-request API bill

06

Learning

Inspect the whole application stack

Local is not automatically private: the surrounding app must avoid cloud calls too.

Pack the model into a smaller suitcase

Store each learned number with fewer bits. The model gets smaller and often faster—with some quality trade-off.

fewer bits → smaller model



Raw weight math is simple. Real memory also includes metadata, context, and runtime buffers.

FP16

16 bits

Largest

Q8

≈ 8 bits

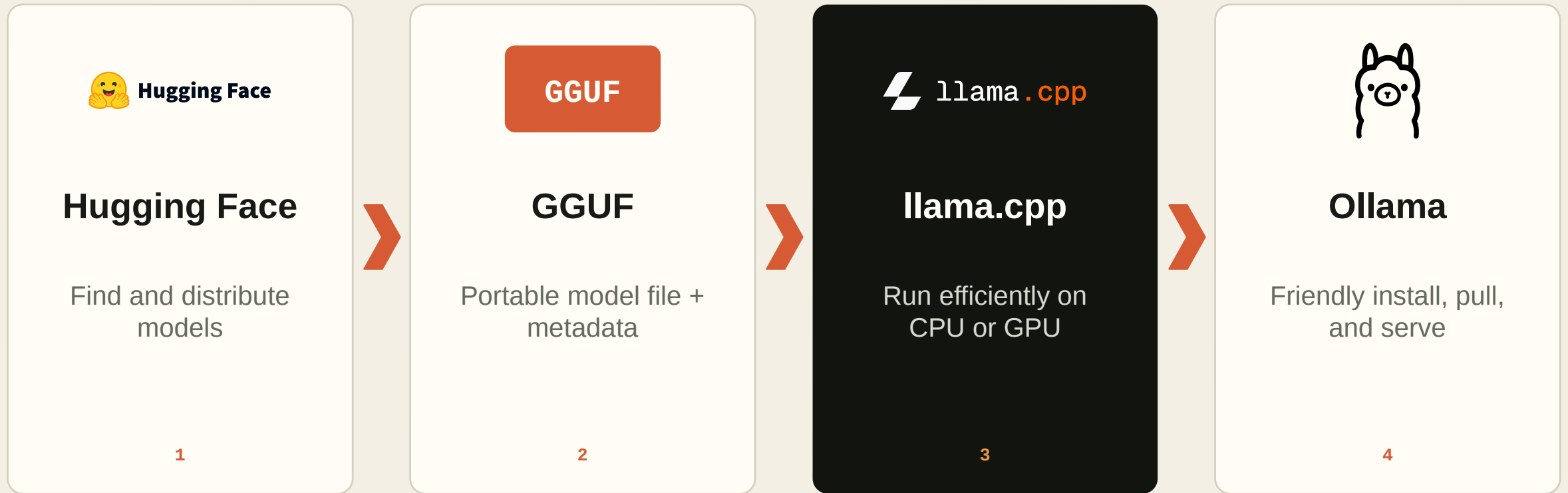
Half-ish

Q4

≈ 4 bits

Quarter-ish

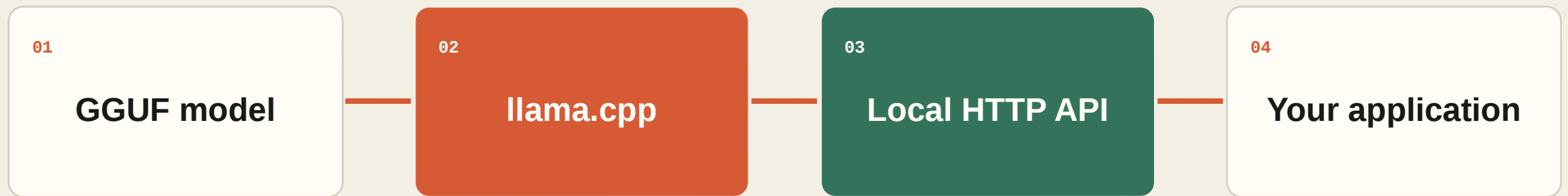
Library → package → engine → easy launcher



An open ecosystem—not one interchangeable license.

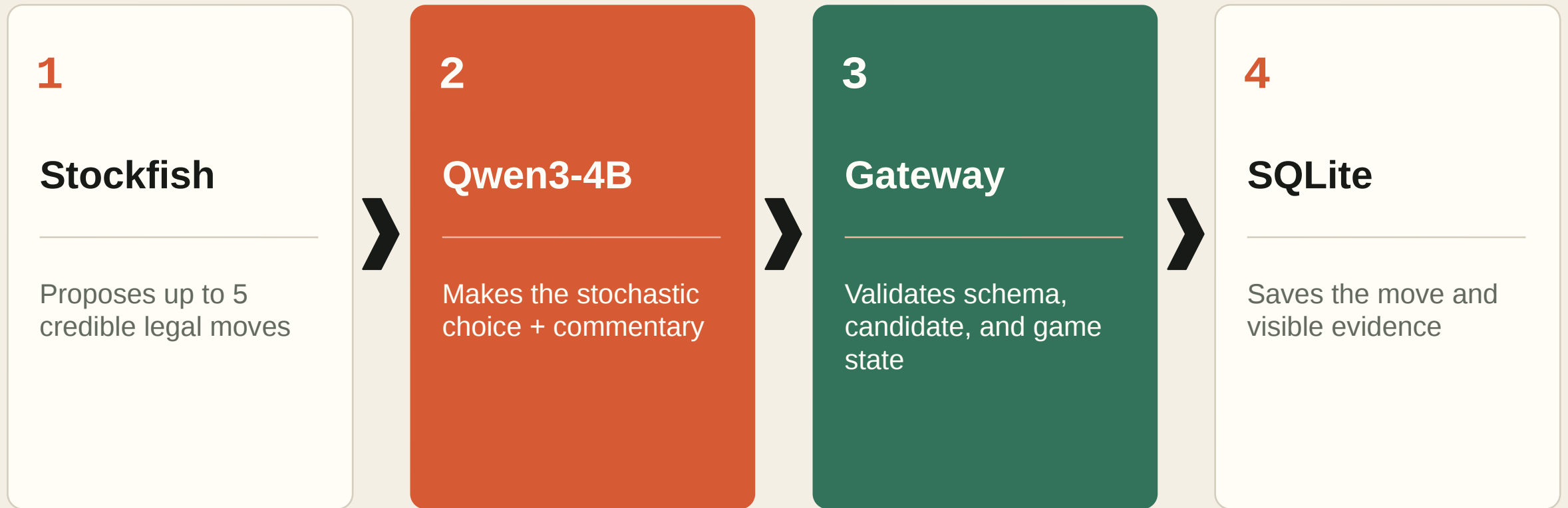
A model file becomes a local service

```
$ llama-server -m model.gguf -ngl 99
```



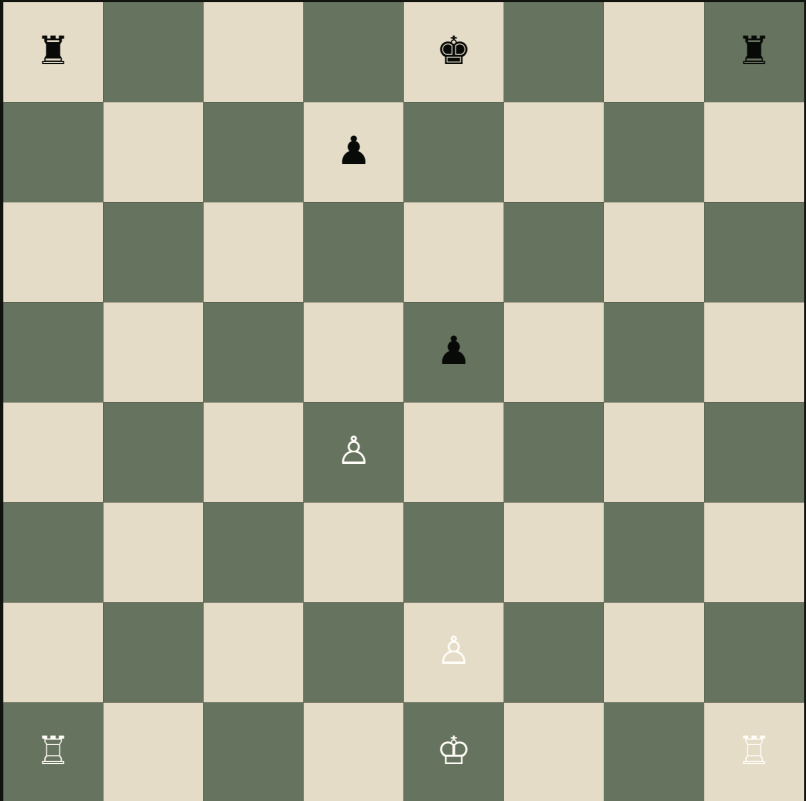
CLI for people. API for programs. The same model stays on your machine.

Constrain creativity. Keep authority in code.



The model chooses every AI move. It cannot invent an illegal one.

Watch one local decision travel through the stack



DECISION TRACE

Stockfish → e4, Nf3, d4

Qwen3-4B → e4

Gateway → valid

Saved → move + evidence

1.8 s · 34 tok/s

RUNNING LOCALLY

- Model: Qwen3-4B
- Quantization: Q4_K_M
- Backend: CUDA
- GPU memory: 8 GB
- Network: not required after setup

WATCH

Candidate moves

Chosen move

Commentary

Latency

Tokens / second

Start with one bounded task

- 01 Use cloud models when maximum capability matters.
- 02 Use local models when privacy, control, latency, or learning matters.
- 03 Build systems that decide what the model may decide.

8

GB

The future is hybrid—
and part of it fits in 8
GB.

Sources

STATE OF AI

Stanford HAI · 2026 AI Index, Technical Performance

HISTORY

Computer History Museum · Timeline of Computer History

MILESTONES

**Google Research · Attention Is All You Need;
OpenAI · Introducing ChatGPT**

LOCAL STACK

**Hugging Face GGUF + Local Apps; llama.cpp
server docs**

HARDWARE

NVIDIA · GeForce RTX 4060 specifications

DEMO

**Chess Llama repository architecture and runtime
manifest**

Full URLs live in slide notes and presentation/README.md.

Image and logo credits

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04 Hugging Face logo · official Hugging Face brand asset

05 llama.cpp logo · ggml-org llama.brand · CC BY-NC 4.0 + nominative-use permission

06 Ollama logo · Ollama repository · MIT license

07 Chess Llama interface preview · original vector composition for this deck

Full URLs, licenses, hashes, and modification notes: [assets/ATTRIBUTION.md](#)

Take the models home

Two practical starting points for building with local models.

REPOSITORY 01



Chess Llama

The local-LLM
chess decision
demo

github.com/kugtong33/labs-chess-llama

REPOSITORY 02



Dockerized llama.cpp

A containerized
local inference
stack

github.com/kugtong33/labs-local-llm